

MAT 127A Discussion 8 Notes

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§I Functional limits

Recall that given a function $f : A \rightarrow \mathbb{R}$ and a limit point c of A , we say that $\lim_{x \rightarrow c} f(x) = L$ provided that, for all $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $x \in A$ with $0 < |x - c| < \delta$, we have $|f(x) - L| < \varepsilon$.

The intuition is that, we have $\lim_{x \rightarrow c} f(x) = L$ if and only if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that after we disregard the values of f at c and anywhere outside of $V_\delta(c)$, the values of f over $V_\delta(c) \setminus \{c\}$ are all contained in $V_\varepsilon(L)$.

(Example 4.2.2) Show that then $\lim_{x \rightarrow 2} (3x + 1) = 7$.

Proof. Let $\varepsilon > 0$. Notice that

$$|(3x + 1) - 7| = |3x - 6| = 3|x - 2|.$$

Thus, if we choose $\delta = \frac{\varepsilon}{3}$, then $0 < |x - 2| < \delta$ implies $|(3x + 1) - 7| < 3\left(\frac{\varepsilon}{3}\right) = \varepsilon$. ■

Remark: Pay attention to how we do not write $|(3x + 1) - 7| = |3x - 6| = 3|x - 2| < \varepsilon$ when making the observation.

1. (Example 4.2.2) Show that $\lim_{x \rightarrow 2} x^2 = 4$.

2. Use all previous results in this discussion to show that $\lim_{x \rightarrow 2} (x^2 + 3x + 1) = 11$.

Proof.

1. A little algebra reveals

$$|x^2 - 4| = |x + 2||x - 2|.$$

Notice how we can control the size of $|x + 2|$ simply by keeping $|x - 2|$ (which is something we can control) small. For precisely, if we have that if $|x - 2| < 1$, then $|x + 2| = |x - 2 + 4| \leq |x - 2| + 4 < 5$.

Let $\varepsilon > 0$, then we can choose $\delta = \min\{1, \frac{\varepsilon}{5}\}$. If $0 < |x - 2| < \delta$, then we have $|x + 2| \leq |x - 2| + 4 < 5$, and so

$$|x^2 - 4| = |x + 2||x - 2| < 5\delta \leq 5\left(\frac{\varepsilon}{5}\right) = \varepsilon.$$

2. Fix $\varepsilon > 0$. Observe that

$$|(x^2 + 3x + 1) - 11| = |(x^2 - 4) + ((3x + 1) - 7)| \leq |x^2 - 4| + |(3x + 1) - 7|.$$

We will make sure that $|x^2 - 4|$ and $|(3x + 1) - 7|$ are both less than $\frac{\varepsilon}{2}$ separately. Since $\lim_{x \rightarrow 2} x^2 = 4$, there exists $\delta_1 > 0$ such that if $0 < |x - 2| < \delta_1$, then $|x^2 - 4| < \frac{\varepsilon}{2}$. Similarly, since $\lim_{x \rightarrow 2} (3x + 1) = 7$, there exists $\delta_2 > 0$ such that if $0 < |x - 2| < \delta_2$, then $|(3x + 1) - 7| < \frac{\varepsilon}{2}$. By setting $\delta = \min\{\delta_1, \delta_2\}$, we have that if $0 < |x - 2| < \delta$, then

$$|(x^2 + 3x + 1) - 11| \leq |x^2 - 4| + |(3x + 1) - 7| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

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Remark: The second part of the previous exercise is a special case of the algebraic limit theorem for functional limits, and its proof is a specialized version of the proof of the algebraic limit theorem.

§II Divergence criterion for functional limits

The direct negation of the definition of functional limits is actually somewhat difficult to work with. Instead, when we are working with a divergent functional limit, it can be more convenient to use sequences.

Given a function $f : A \rightarrow \mathbb{R}$ and a limit point c of A , show that $\lim_{x \rightarrow c} f(x) = L$ implies that all sequences (x_n) in $A \setminus \{c\}$ with $x_n \rightarrow c$ satisfy $f(x_n) \rightarrow L$.

Proof. Suppose that $\lim_{x \rightarrow c} f(x) = L$ and (x_n) is a sequence in $A \setminus \{c\}$ with $x_n \rightarrow c$. Let $\varepsilon > 0$, then we wish to show that there exists $N \in \mathbb{N}$ such that for all $n \geq N$, we have $|f(x_n) - L| < \varepsilon$. Since $\lim_{x \rightarrow c} f(x) = L$, we can control the size of $|f(x_n) - L|$ by controlling the size of $|x_n - c|$, which we can control since $x_n \rightarrow c$. More precisely, there exists $\delta > 0$ such that

$$0 < |x - c| < \delta \quad \text{implies} \quad |f(x) - L| < \varepsilon;$$

and there exists $N \in \mathbb{N}$ such that

$$n \geq N \quad \text{implies} \quad |x_n - c| < \delta.$$

To verify that this N works, let $n \geq N$. Then we have $|x_n - c| < \delta$ and because $x_n \neq c$, we have $0 < |x_n - c| < \delta$, which implies $|f(x_n) - L| < \varepsilon$. ■

A direct consequence of the previous exercise is the following divergence criterion for functional limits:

Let $f : A \rightarrow \mathbb{R}$ and assume that c is a limit point of A . If there exist two sequences (x_n) and (y_n) in $A \setminus \{c\}$ such that

$$\lim x_n = \lim y_n = c \quad \text{but} \quad \lim f(x_n) \neq \lim f(y_n),$$

then $\lim_{x \rightarrow c} f(x)$ does not exist.

Come up with and give brief justifications for each of the following requests.

1. A bounded function $f : \mathbb{R} \rightarrow \mathbb{R}$ whose functional limits $\lim_{x \rightarrow c} f(x)$ exists for all $c \neq 0$ but fails to exist at $c = 0$.
2. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ whose functional limits $\lim_{x \rightarrow c} f(x)$ only exists when $c = 0$.

Proof.

1. Consider the function

$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0. \end{cases}$$

For $c \neq 0$, if $c > 0$, then we can easily see that $\lim_{x \rightarrow c} f(x) = 1$ (for every $\varepsilon > 0$, we can choose $\delta = \frac{\varepsilon}{2}$). Similarly, if $c < 0$, then we have $\lim_{x \rightarrow c} f(x) = -1$. However, at $c = 0$, we can consider

the sequences $x_n = \frac{1}{n}$ and $y_n = -\frac{1}{n}$, both of which converge to 0. However, we have $f(x_n) = 1$ for all n , so $\lim f(x_n) = 1$; whereas we have $f(y_n) = -1$ for all n , so $\lim f(y_n) = -1$. Thus, by the divergence criterion, $\lim_{x \rightarrow 0} f(x)$ does not exist.

2. Consider the function

$$f(x) = \begin{cases} x & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

For $c = 0$, we have $\lim_{x \rightarrow 0} f(x) = 0$ because for every $\varepsilon > 0$, we can choose $\delta = \varepsilon$, and then for all x with $0 < |x - 0| < \delta$, we have

$$\begin{aligned} |f(x) - 0| &= x < \delta = \varepsilon & \text{if } x \in \mathbb{Q} \text{ and} \\ |f(x) - 0| &= 0 < \varepsilon & \text{if } x \notin \mathbb{Q}. \end{aligned}$$

For $c \neq 0$, we can pick $a_n \in \mathbb{Q} \cap V_{1/n}(c)$ and $b_n \in (\mathbb{R} \setminus \mathbb{Q}) \cap V_{1/n}(c)$ for each $n \in \mathbb{N}$ to form by the density of rationals and irrationals. Then both sequences (a_n) and (b_n) converge to c . However, we have $\lim f(a_n) = \lim a_n = c$ and $\lim f(b_n) = \lim 0 = 0$. Thus, by the divergence criterion, $\lim_{x \rightarrow c} f(x)$ does not exist for any $c \neq 0$.

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